

IMPERIAL

External flow around an aircraft

CFD Project Assignment 2

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Agenda

- 1 Introduction
- 2 Flow definition
- 3 Turbulence models
- 4 Grid definition and refinement
- 5 Take-off speed analysis
- 6 Angle of Attack effects analysis
- 7 Conclusions and Q&A

Introduction

Case definition

Scenario

Case study overview

AIRCRAFT

- Lockheed **C-130 Hercules**
 - Four Engine turboprop for military use
- Aerodynamic interest:
 - High wings & large fuselage CSA
 - Low cruising speed



Figure 1 – Lockheed C-130 Hercules

GEOMETRY AND MODELS

- **Half domain** modelled exploiting symmetry:
 - Reduced computational costs
- **Turbulence models** employed:
 - Standard k- ϵ two-layer; all- y^+ treatment
 - SST k- ω
 - Quadratic pressure strain Reynolds stress; high- y^+ wall treatment

OBJECTIVES

- Gain insights into the lift and drag characteristic of the aircraft:
 - Determine **minimum take-off speed**
 - Investigate the **effects of angle of attack** on flow field and integral quantities (during cruising)

Dimensionless numbers

Dimensionless numbers defining the problem

LIFT COEFFICIENT

Lift-generating capability

$$C_L = \frac{L}{\frac{1}{2} \rho_{\infty} V_{\infty}^2 A_{C_L}}$$

DRAG COEFFICIENT

Drag-generating capability

$$C_D = \frac{D}{\frac{1}{2} \rho_{\infty} V_{\infty}^2 A_{C_D}}$$

PRESSURE COEFFICIENT

Local P deviation from freestream

$$C_p = \frac{p - p_{\infty}}{\frac{1}{2} \rho V_{\infty}^2}$$

REYNOLDS NUMBER

Ratio between inertia and viscous forces

$$Re = \frac{\rho V_{\infty} x}{\mu_{\infty}}$$

MACH NUMBER

Ratio of fluid velocity to sonic speed

$$M = \frac{V_{\infty}}{c}$$

Flow Definition

Simulation Setup

Geometry & boundary conditions

GEOMETRY

Blockage factors limited to maximum 10%¹:

$$\text{blockage factor} = A_{\text{object}} / A_{\text{wind tunnel}}$$

Table 1 – Geometries dimensions

	Height (H)	Width (W)	Length (L)
C-130	4.57 m	4.36 m	29 m
Wind tunnel	10 H	3.5 W	8 L



- Maximum *blockage factor* of **2.89%**
- Aircraft in the **middle** of wind tunnel height and **2L-3L** along length

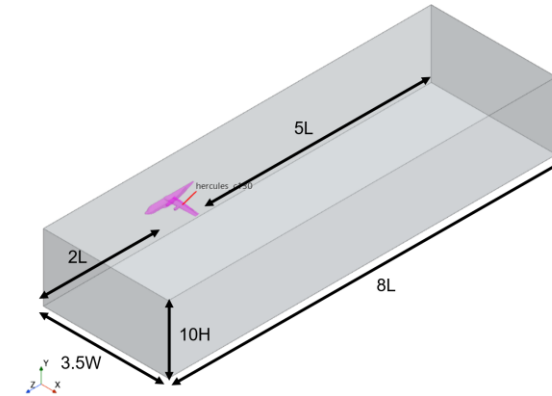


Figure 2 – Wind tunnel geometry

BOUNDARY CONDITIONS

- Taken at **2200 feet & 180 m/s**
- Walls conditions:
 - Top, bottom, X=3.5W: **slip-walls**
 - Front: **velocity** inlet (180 m/s)
 - Back: **pressure** outlet (42,789 Pa)
 - X=0: **symmetry**
- $Re = 2.44e7$

Turbulence Models

Background Theory

MODELLING TURBULENCE

1. Velocity Decomposition

$$u_i = U_i + u'_i$$

2. RANS Momentum Equation

$$\frac{\partial U_i}{\partial t} + U_j \frac{\partial U_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial P}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\nu \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right) - \overline{u'_i u'_j} \right]$$

3. How to close the system

- Two-equation Model: Boussinesq Approximation

$$\overline{u'_i u'_j} = \overline{u'v'} = \nu_t \left(\frac{\partial U}{\partial y} + \frac{\partial V}{\partial x} \right)$$

- Second-Moment closure model: No approximation

WALL FUNCTION

1. Y^+ : Size of the first element

2. Near – Walls Flows:

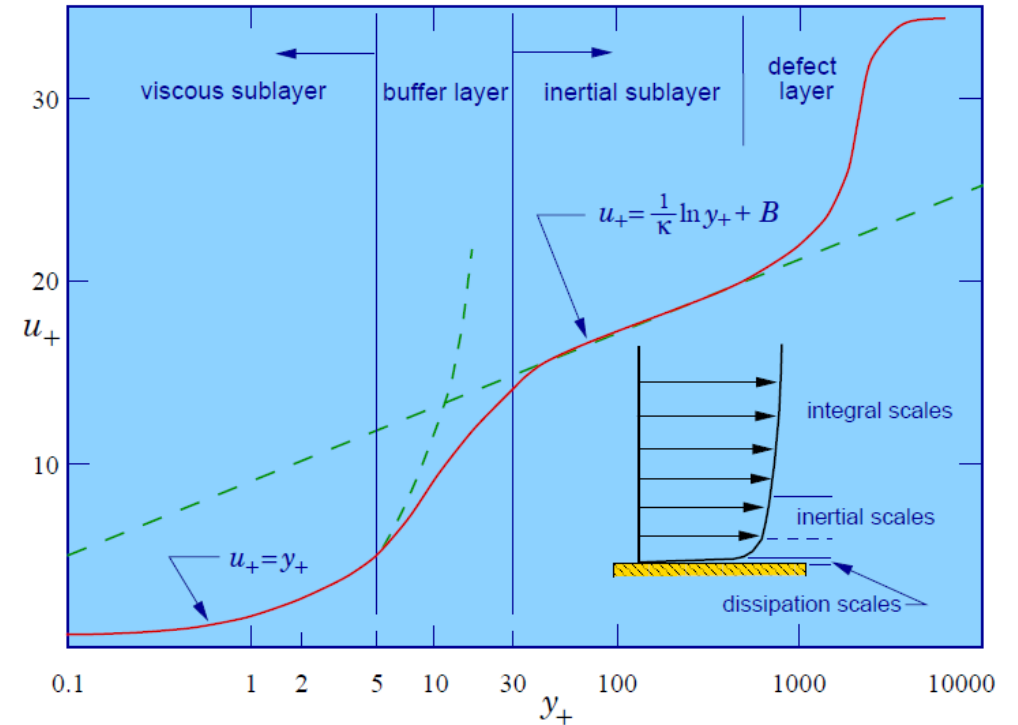


Figure 3 – Near-wall flow diagram¹

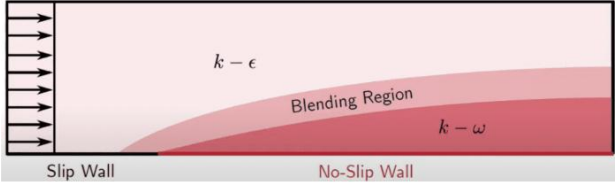
The three models

k-ε, k-ω, Reynolds Stress

How does it close the system of equations?

Which Version

Wall Treatment

k-ε	k-ω	REYNOLDS STRESS
Two new transport equations for k and ε $\nu_t = c_\mu \frac{k^2}{\varepsilon}$	Two new transport equations for k and ω $\nu_t = \frac{k}{\omega}$	Six new transport equations for every single component of the Reynolds stress tensor
Standard k-ε Two-Layer ➤ Outer Layer: k and ε equations ➤ Inner layer: k equation, ε and ν_t algebraically	Shear stress transport (SST)  Figure 4 – SST k-ω definition¹	Quadratic Pressure Strain ➤ Quadratic formulation for the pressure strain component
All Y+ Treatment ➤ Dynamically adapts the wall formulation based on the local Y+ value of the first mesh cell	All Y+ Treatment ➤ Dynamically adapts the wall formulation based on the local Y+ value of the first mesh cell	High Y+ Treatment ➤ Applies log-law function instead of solve the viscous sublayer

Grid Definition & Refinement

Grid definition

Parameters and features of the mesh

Table 2 – Parameters for grid definition

Parameter	Description
Base Size	- Fundamental reference length - Most of other parameters defined as % of base size
Maximum Cell Size	- Largest allowable size % of Base
Prism Layer Thickness	- Thickness of layer of prismatic cells growing outward from aircraft - Essential for boundary layers
Relative Minimum Size (Surface Size)	- Smallest size permitted on a specific region (surface) % of Base
Relative Target Size (Surface Size)	- Desired size on a specific region (surface) % of Base

MESH FEATURES

- Total cell count: **1,801,489**
- Contact prevention:
 - Between flaps and main body
 - Minimum size: 0.001 m
- Volumetric Controls:
 - A) Landing gear body
 - B) Wing and fuselage connection

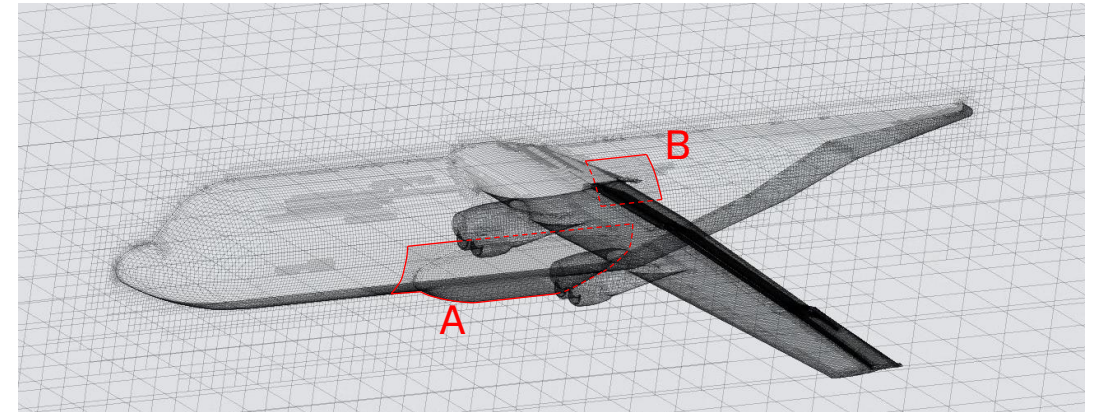


Figure 5 – Fuselage mesh with highlighted volumetric controls

Grid refinement

Integral quantity independence

METHODS

- Identify 3 grids with **increasing cell count**
- Determine C_L and C_D for each grid using the **three turbulence models**
- Assess the gradient between grids to determine **independence (flat gradient)**

Table 3 – Reference properties for C_L and C_D

Parameter	Property	Value	Units
C_L, C_D	Reference Pressure	-68,325	Pa
C_L, C_D	Reference Density	0.49	kg/m ³
C_L, C_D	Reference Velocity	180	m/s
C_L	Reference Area	81.05	m ²
C_D	Reference Area	80.05	m ²

C_L AND C_D GRID-INDEPENDENCE

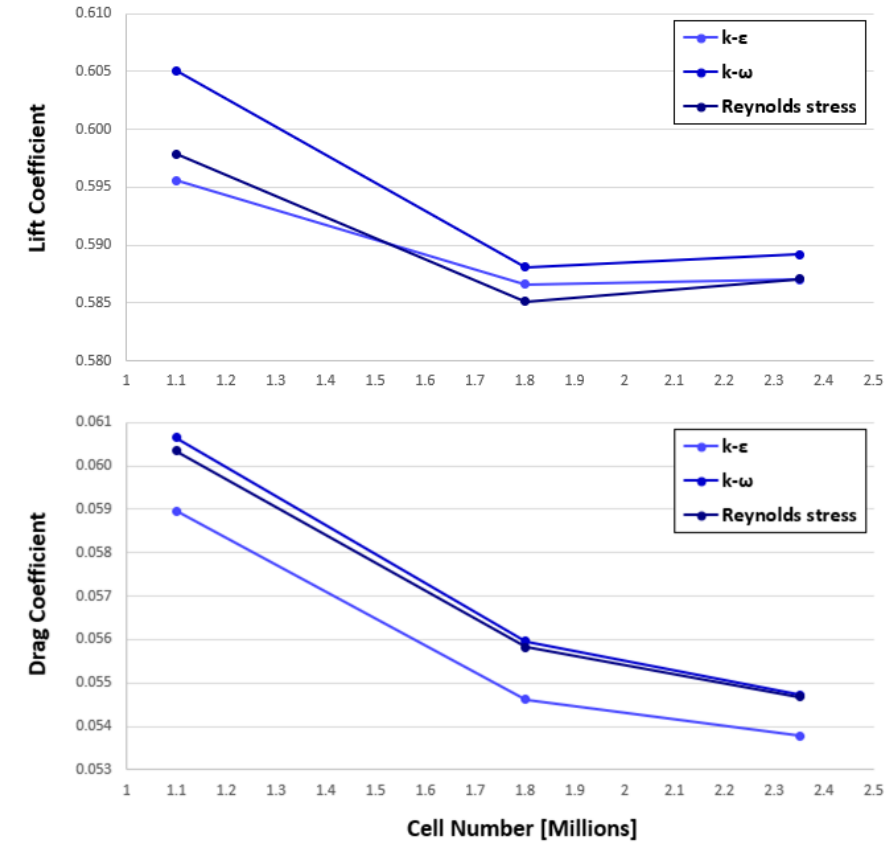


Figure 6 – C_L and C_D evolution with cell count

Reference grid – Discussion 1/4

Refinement impact on integral quantities

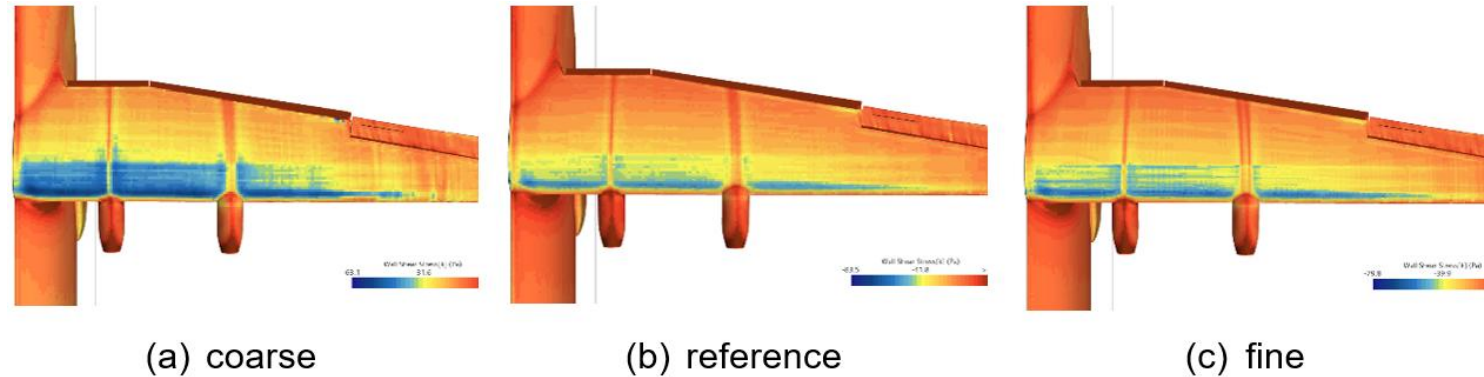


Figure 7 – Wall Shear Stress contour plots for three different grid resolution: a) coarse, b) reference, c) fine

GRIDS EFFECTS

- Excessive **numerical dissipation** in coarse grid shown by “blocky” distribution
- Shear stress minimum **underestimated** for coarse grid (~ -63 Pa) and **stable** (~ -80 Pa) for reference and fine grids
- **Minimal refinements** from reference to fine

SOLUTIONS GRADIENTS EVOLUION

As seen in Figure 6, relative rate of change:

- Coarse to reference:
 - C_L : 2.12%
 - C_D : 3.32%
- Reference to fine:
 - C_L : 0.33%
 - C_D : 0.37%

Reference grid – Discussion 2/4

Grid feature analysis

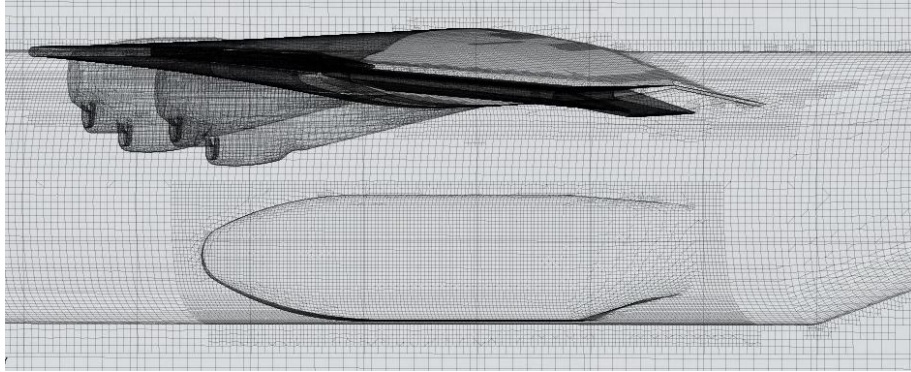


Figure 8 – Zoom in mesh control volumes

Simcenter STAR-CCM+



Figure 9 – Velocity field around fuselage extrusions

Simcenter STAR-CCM+

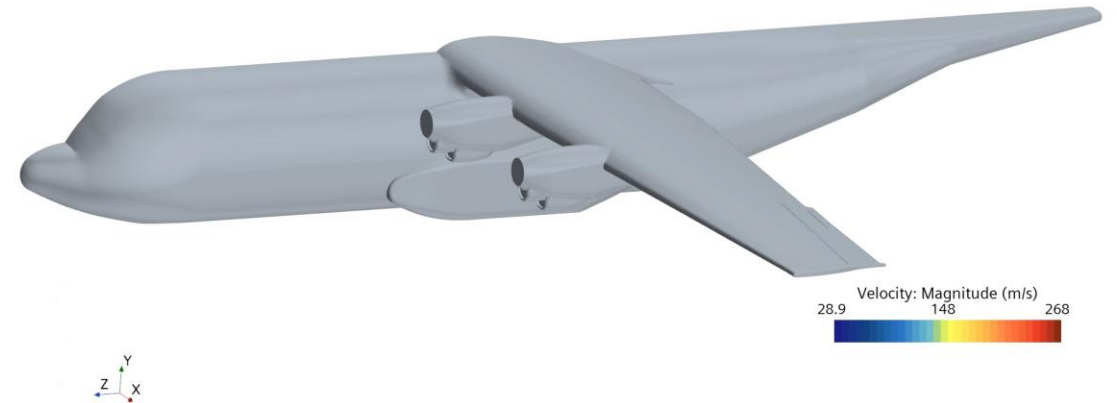


Figure 10 – Velocity field around aircraft

Reference grid – Discussion 3/4

Y^+ Layer

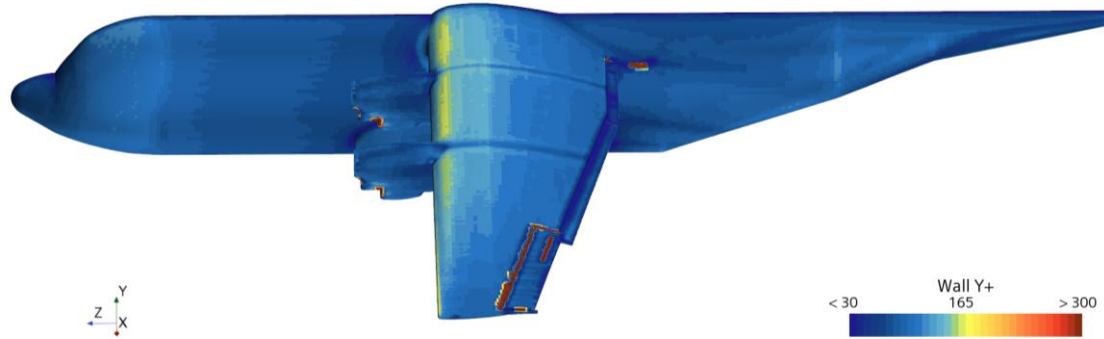


Figure 11 – Wall Y^+ scalar contour

Y^+ VALUES ANALYSIS

- **First cell in the log-law** region (no wall-integrated approaches)
- Allowed region: **$30 < y^+ < 300$** (Log-law layer)*
- Regions where **geometry impedes target y^+** :
 - Increase number of cells
 - Acknowledge limitation & check impact

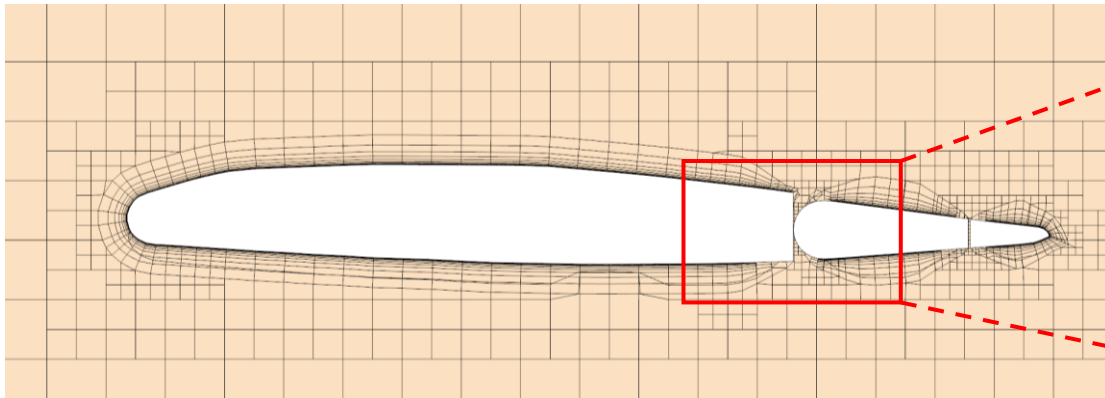


Figure 12 – Mesh grid section at outer flap-wing intersection

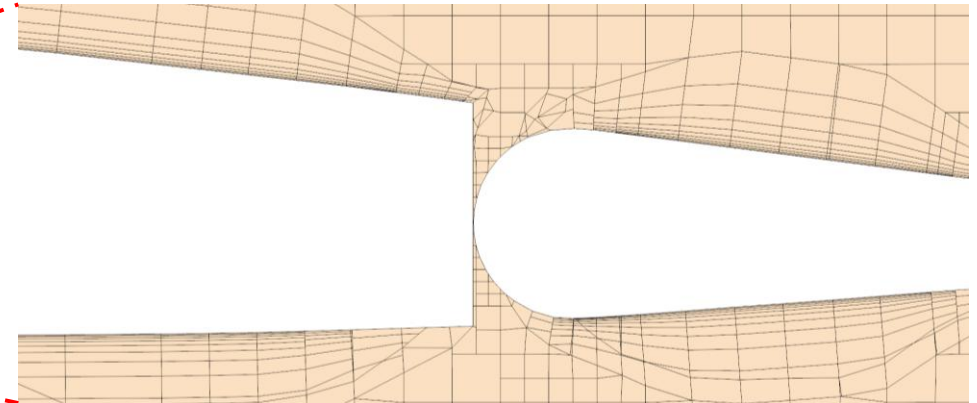


Figure 13 – Zoom in the region with sharp geometry

Reference grid – Discussion 4/4

Quality of solution

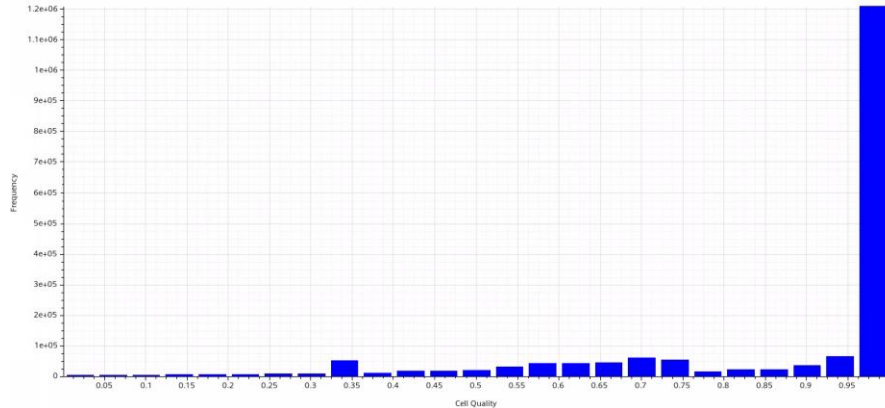


Figure 14 – Cell quality histogram

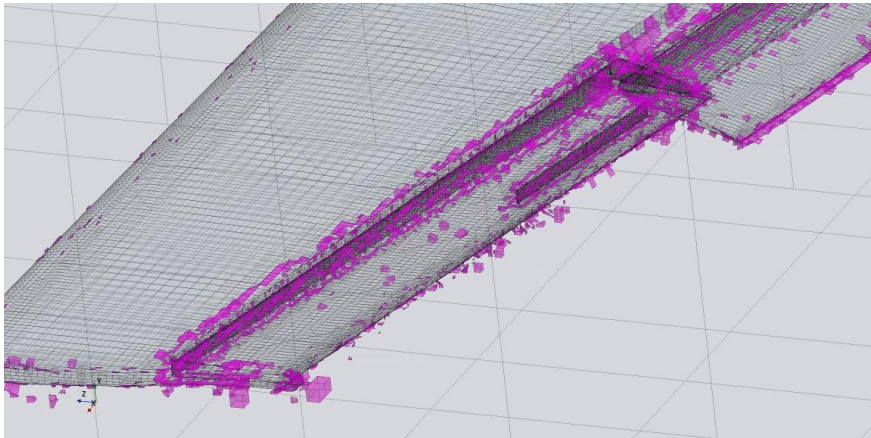


Figure 15 – Wing section with highlighted low-quality cells

CELL QUALITY ANALYSIS

- StarCCM+ embedded cell quality report assigns score of 0 to degenerate cells and score of 1 to perfect cells, considering:
 - Cell Volume relative to ideal
 - Aspect Ratio
 - Face planarity
 - Skewness
- Criterion: **<~1% of low-quality cells** (<0.2)
- Mainly cells in “perfect” region ($>90\%$ in $0.9+$)
- **Clusters of low-quality** in regions with sharp geometries and narrow spaces (highlighted in purple)

Take-off Speed Analysis

Take-off speed analysis

METHODS

- Maximum take-off weight: 74393 kg
- Empty weight: 40030 kg
- Velocity inlet boundary condition varies until lift force on model equals aircraft weight

MODIFICATIONS TO SYSTEM

- Boundary condition for bottom plane: no-slip
- Angle of attack for take-off: 5°
- New aircraft location: 0.5 m above floor

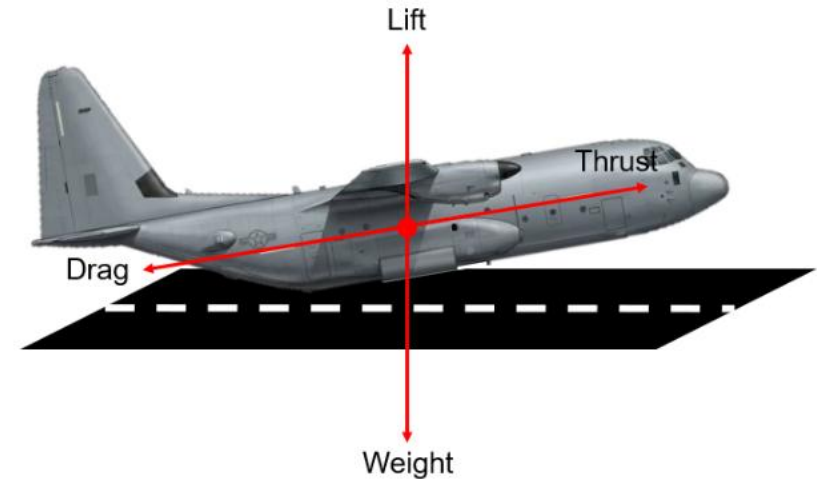


Figure 16 – Free Body Diagram of the plane (take-off)

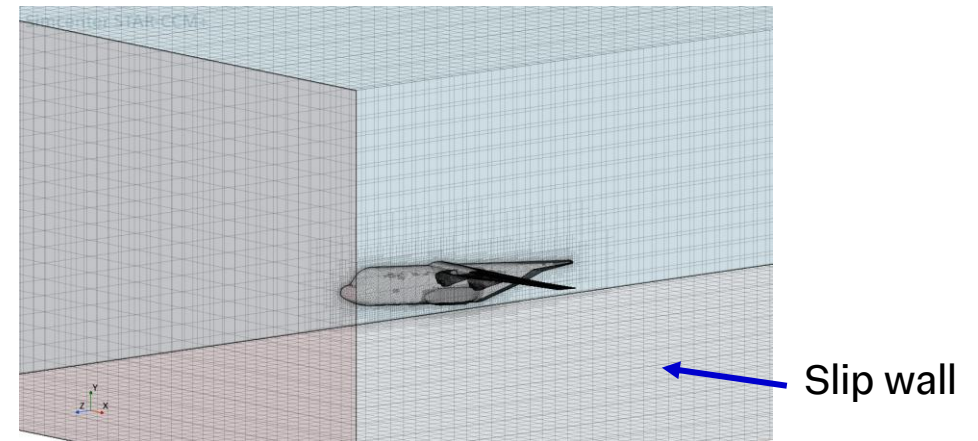


Figure 17 – Mesh with aircraft for take-off analysis

Take-off speed analysis

Table 4 – Take-speed for different models and conditions

Turbulence model	Maximum payload estimated speed	Maximum payload estimated speed
k-ϵ	83.3 m/s	63.8 m/s
k-ω	83.9 m/s	64.1 m/s
RSM	83.2 m/s	62.1 m/s

- Literature take-off: ~120 knots¹ (62 m/s)
- Reasons for discrepancies:
 - Engine not modelled → no thrust effect
 - Lacking tail-wing and landing gears

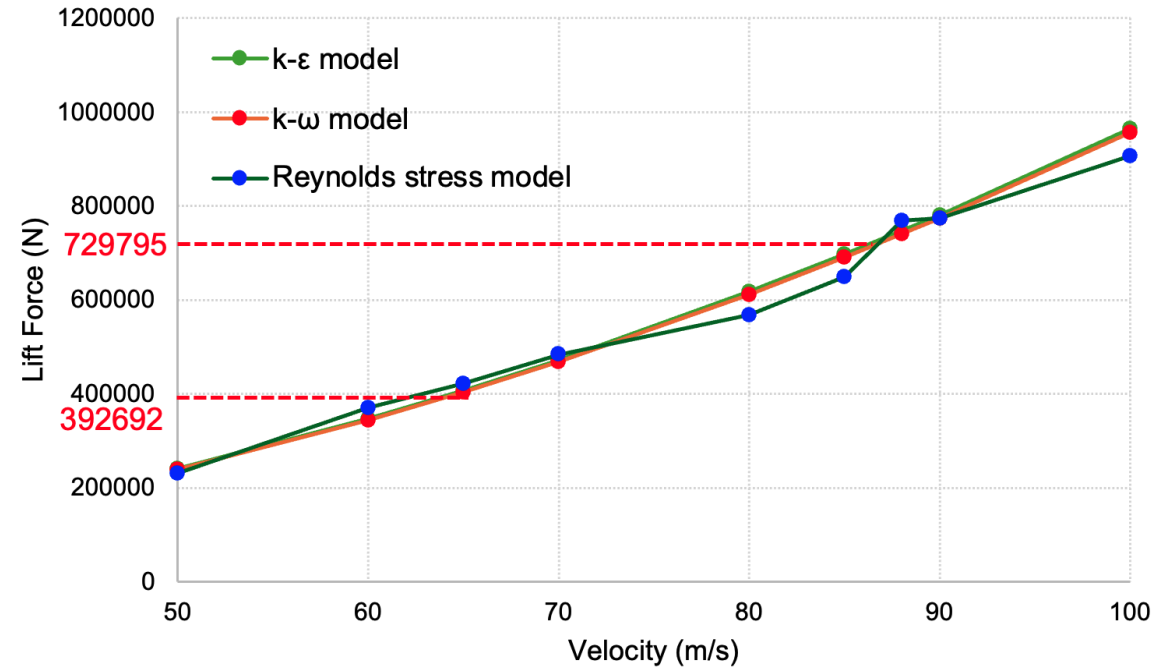


Figure 18 – Lift force generated vs. aircraft speed

Angle of Attack effects analysis

Angle of Attack analysis

Goal and methods of the study

OBJECTIVES

- Assess the influence of increasing AoA on the aircraft under fixed cruise conditions
- Compare the performance of the three turbulence models (k- ϵ , SST k- ω , RSM)
- Identify the angle of separation onset

METHODS

Quantities investigated:

- C_L
- C_D
- C_p plotted against x/c

Angles:

- Initial sweep: 5° 10° 12.5° 15° 17.5° 20°
- Finer range around peak C_L

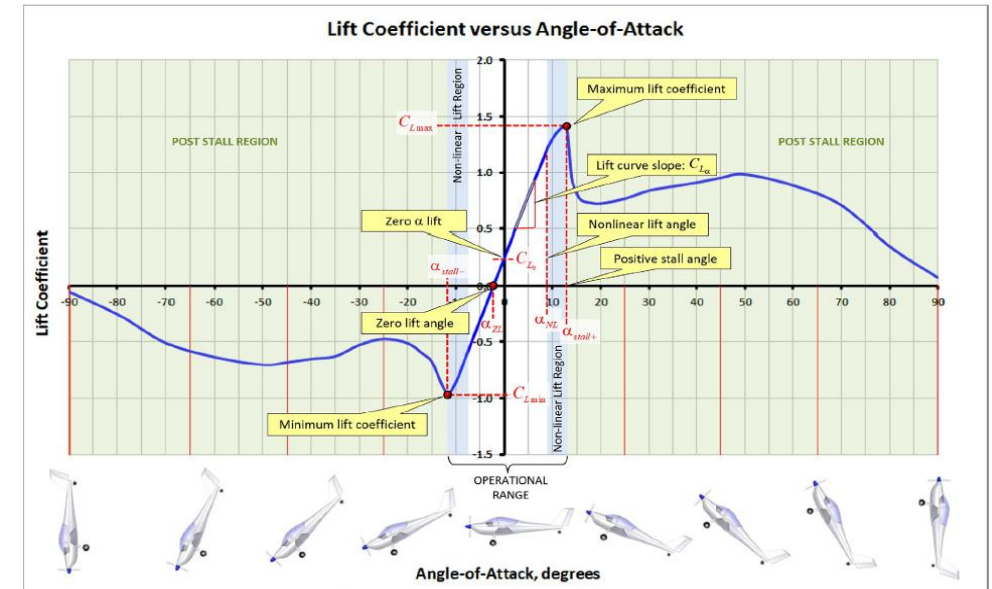


Figure 19 – C_L vs. AoA literature analysis¹

SOLUTION EXPECTED BEHAVIOUR

- C_L increases almost linearly until boundary layer separation
- At separation: max C_L reached
- Beyond separation: “stall”, C_L decreases

Angle of Attack – Discussion 1/4

Lift Coefficient

LIFT COEFFICIENT EVOLUTION

C_L varies with model and AoA:

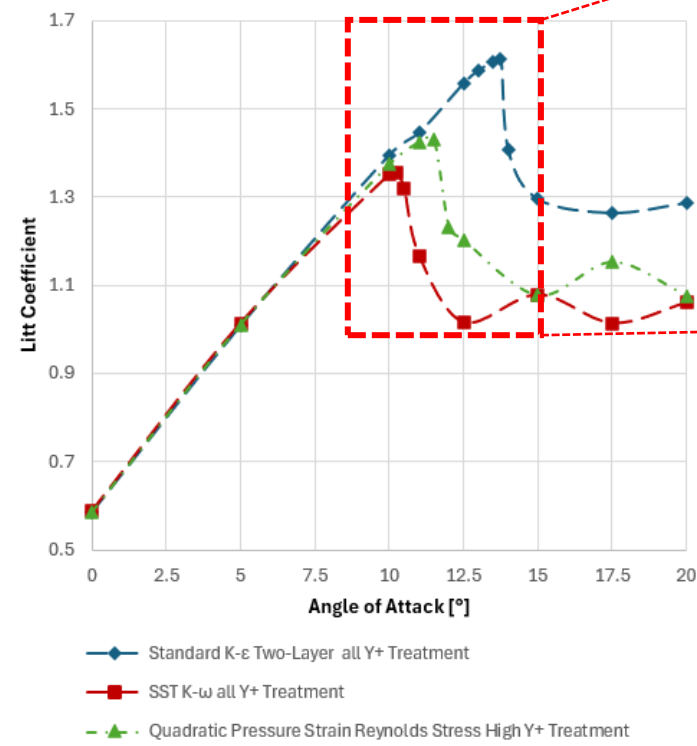


Figure 20 – C_L vs. aircraft angle of attack

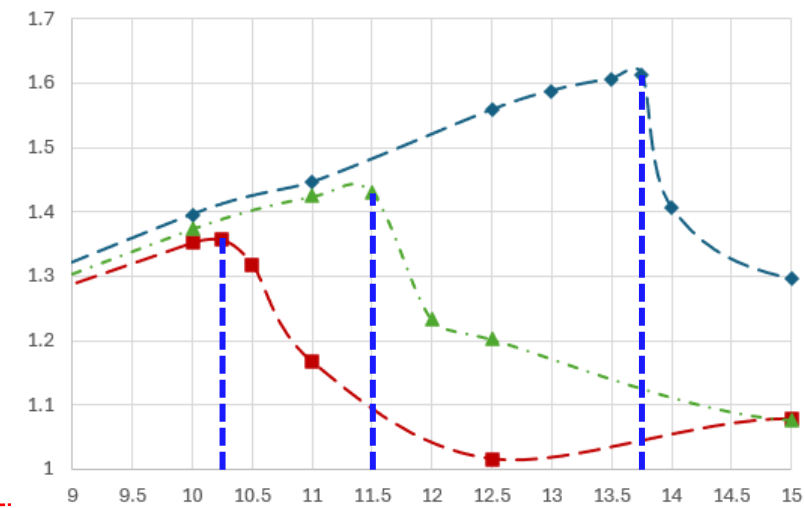


Figure 21 – C_L vs. aircraft angle of attack, zoom in

Table 5 – Take-speed for different models and conditions

Turbulence model	AoA at maximum C_L
k-ε	13.75°
k-ω	10.25°
RSM	11.5°

Angle of Attack – Discussion 2/4

MAC and flow separation effects on C_D

MEAN AERODYNAMIC CHORD

- Different cross sections produce different flow separation behaviours
- Mean Aerodynamic Chord: removes local-flow effects (average aerodynamic behaviour)
- $y_{\bar{c}} = 9.98 \text{ m}$

FLOW SEPARATION EFFECTS ON C_D

- **Trend:**
 - Smooth increase until flow separation
 - Sharp rise when separation occurs
 - Smooth increase after separation
- **Rationale:** flow separation creates turbulent low-pressure wake at rear → sudden increase in $\Delta P_{\text{rear-front}}$ → sharp increase in C_D

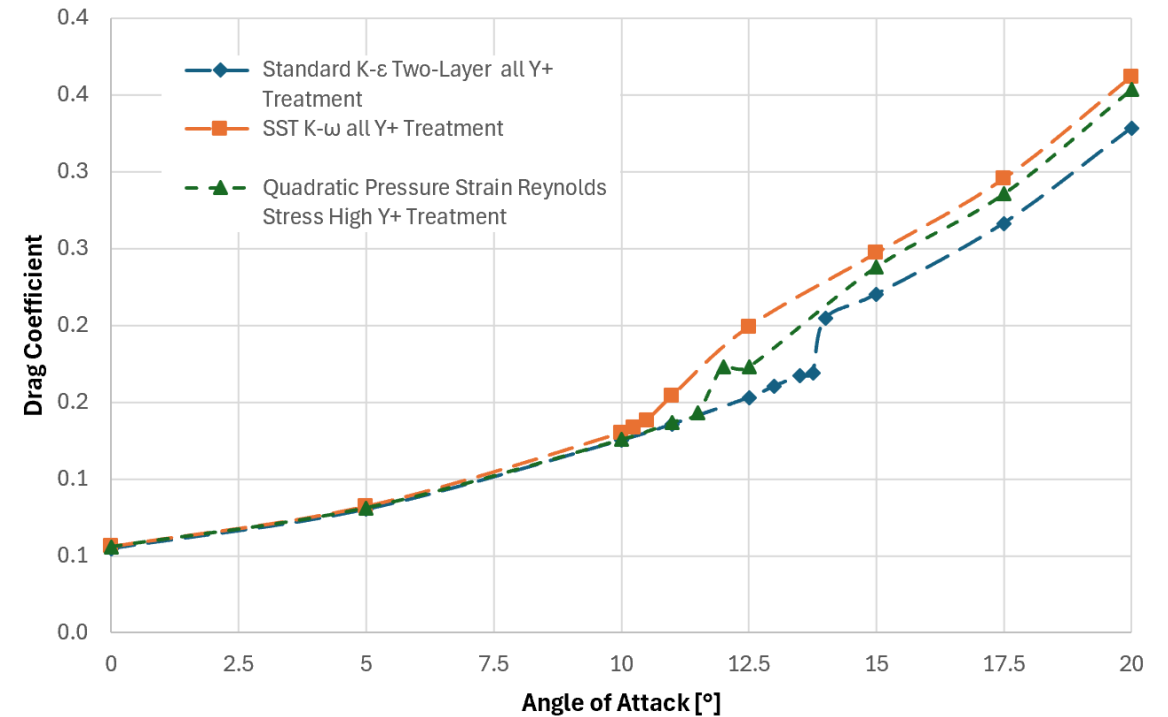


Figure 22 – C_D vs. aircraft angle of attack

Angle of Attack – Discussion 3/4

Flow separation effects on velocity and C_p

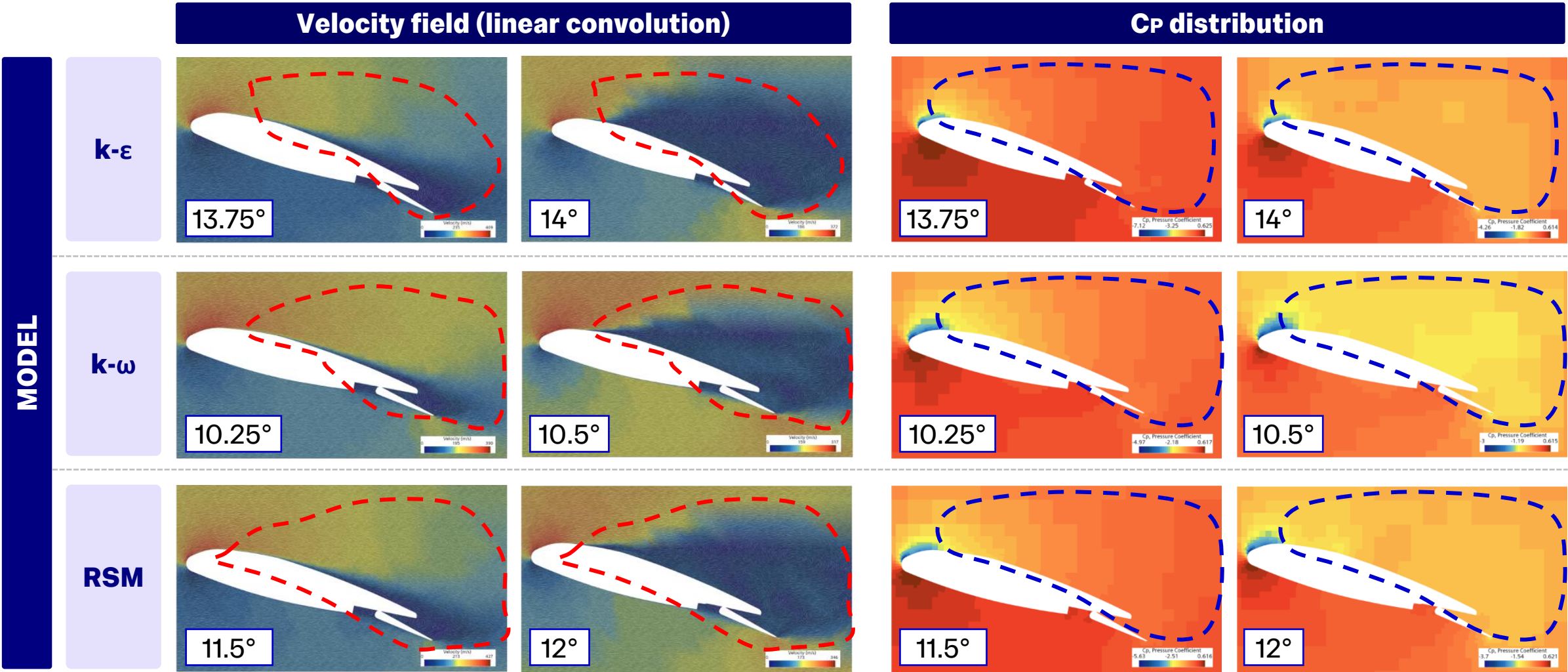


Figure 23 – velocity field and C_p distribution for the three models at flow separation AoA

Angle of Attack – Discussion 4/4

Flow separation effects on C_P

FLOW SEPARATION EFFECTS ON C_P

- Peak C_P at the leading edge due to max suction, decreasing toward the trailing edge
- $|C_P|$ before separation is higher until $x/c = 0.65$ (attached flow accelerates more over the surface) but lower beyond 0.65 (likely due to spanwise flow and tip vortices altering boundary layer momentum)

Table 6 – Peak C_P values and corresponding x/c

Turbulence model	x/c	Maximum C_P
k-ϵ	0.01	-7.12
k-ω	0.01	-4.97
RSM	0.00	-5.64

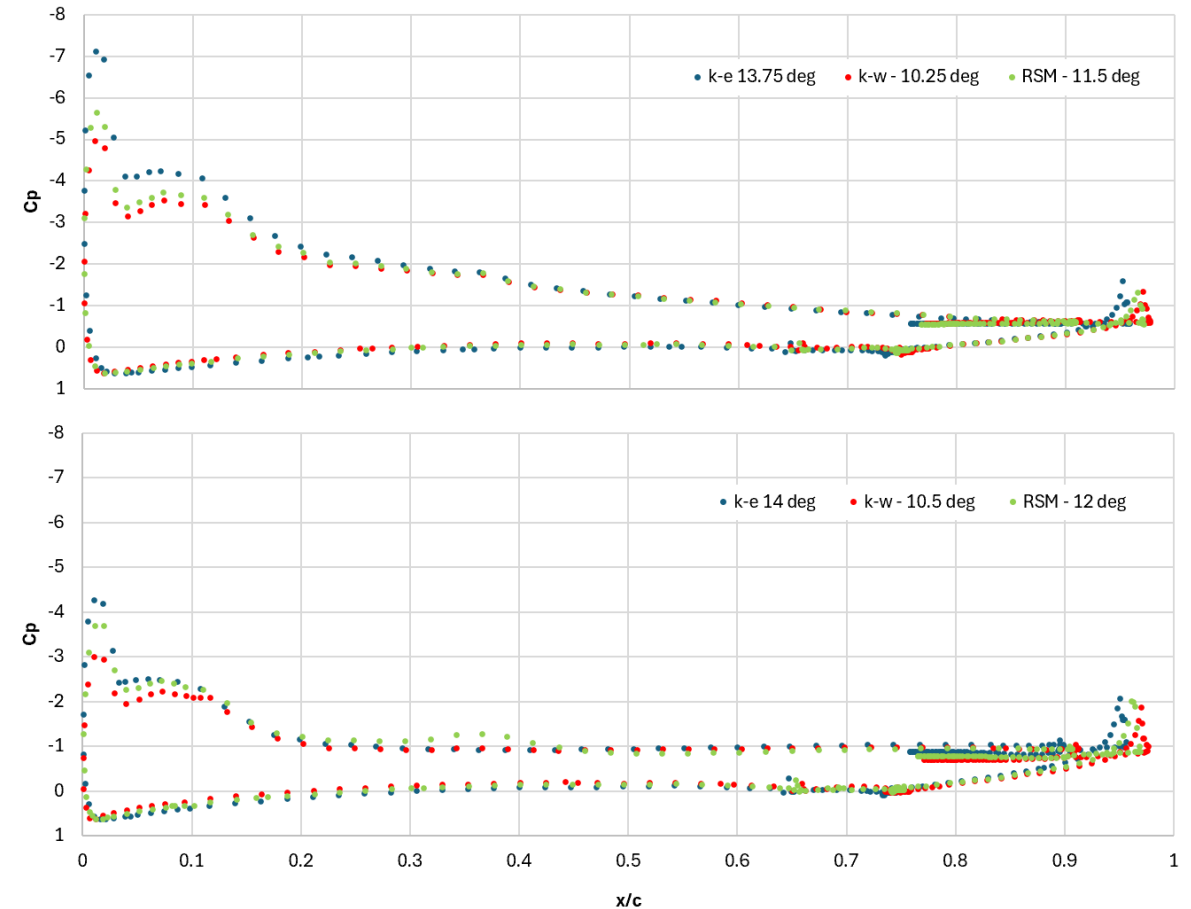


Figure 24 – C_P along the chord of the wing

Conclusions

Summary

Key findings from the CFD analysis of the C-130 Hercules

GRID REFINEMENT

- Reference mesh: **1,801,489 cells**
- CL change confirms **grid independence**:
 - coarse → reference: 2.12%
 - reference → fine: 0.33%
- Cell quality: **>90% in perfect region**; low-quality cells isolated at sharp geometry features

ANGLE OF ATTACK – C_L & C_D

- **Separation onsets at maximum C_L :**
 - $k-\epsilon$: 13.75°
 - $k-\omega$: 10.25°
 - RSM: 11.5°
- $k-\epsilon$ predicts delayed stall; $k-\omega$ earliest separation
- **C_D increases monotonically** with AoA and the three models diverge post-separation

TAKE-OFF SPEED ANALYSIS

- All **three models agree** closely at max payload: $k-\epsilon$: 83.3 m/s, $k-\omega$: 83.9 m/s, RSM: 83.2 m/s
- Discrepancies with literature due to **absent engines, landing gear, and tail wing**
- Turbulence model choice has minimal impact on take-off speed prediction

ANGLE OF ATTACK – C_p & FLOW

- **Peak suction at leading edge** ($x/c \approx 0.01$) and $|C_p|$ decreases toward trailing edge
- Velocity fields confirm boundary layer detachment and vortex formation at separation AoA for all models

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Thank you
Q&A session now

16/03/2026